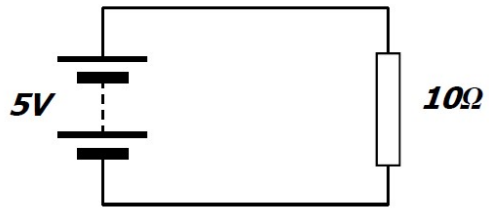
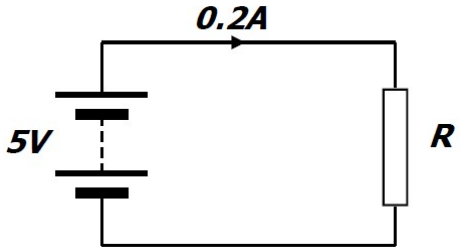
Analogue Electronics Worksheet

1. Calculate the missing quantity in the following circuits.

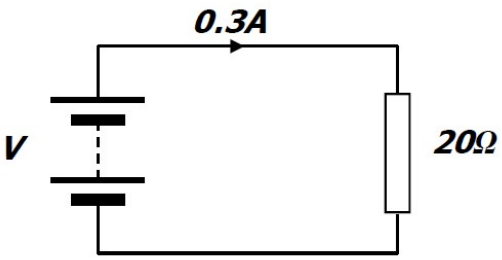
   a. Find the current.



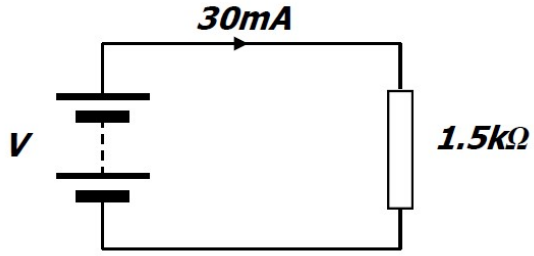
   b. Find the resistance of the resistor, R.



   c. Find the battery voltage.

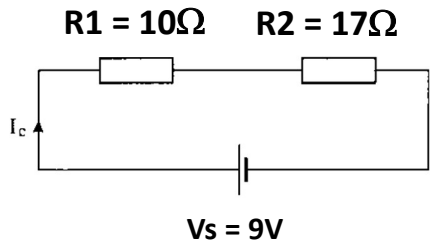


   d. Find the battery voltage.

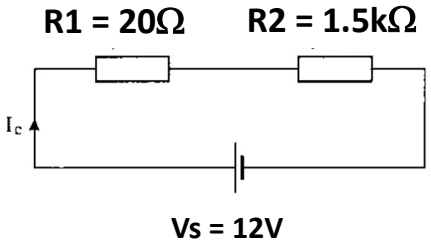


2. In your jotter, draw the following series circuits and calculate:
   a) The total resistance (R_T).
   b) The circuit current (I_C). (I = V/R)
   c) The voltage across each resistor. (V = IR)

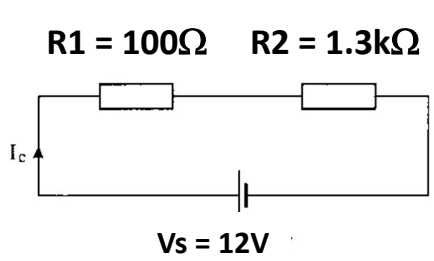
a)



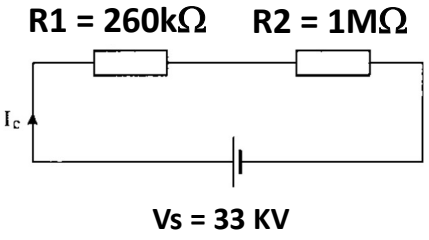
b)



c)

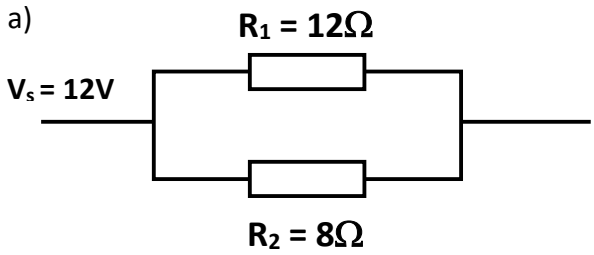


d)

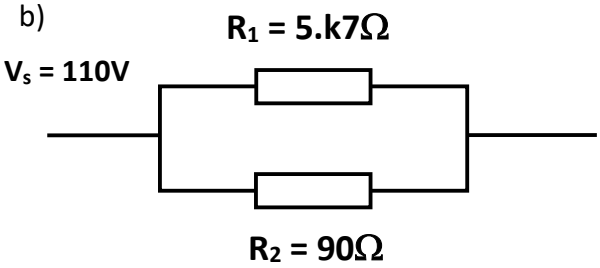


3. In your jotter, draw the following parallel circuits and calculate:
   a) The total resistance (R_T).
   b) The circuit current (I_C). ($I_c = V_s/R_T$)
   c) The current across each resistor (I_1 and I_2 and so on).

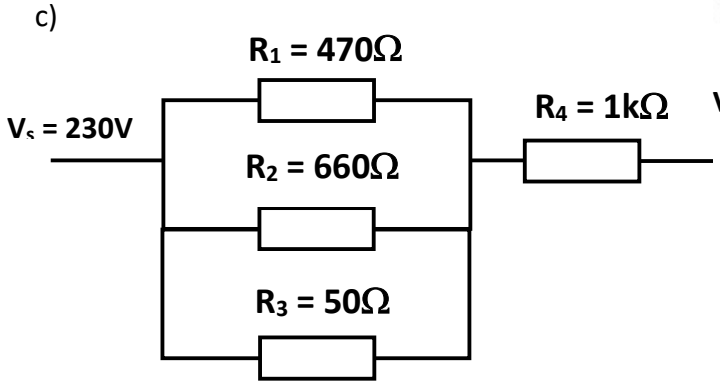
a)



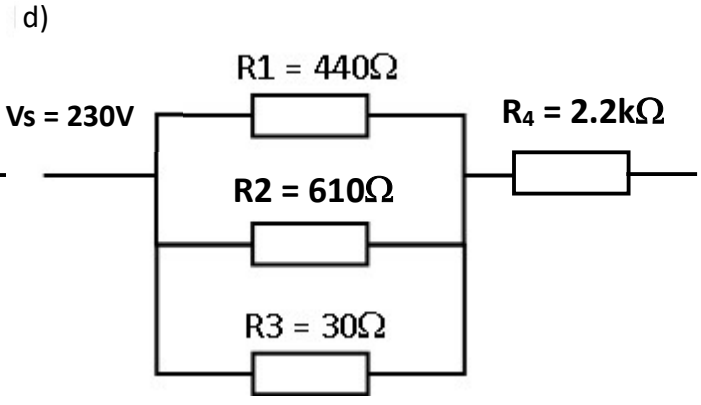
b)



c)



d)



4. Two resistors (15kΩ, 25kΩ) are connected in parallel and then connected to a 6v supply voltage. Calculate;
   a) The total resistance.
   b) The circuit current.

5. Three resistors (2kΩ, 5kΩ, and 10kΩ) are connected in parallel and then connected to a 10v supply voltage. Calculate;
   a) The total resistance.
   b) The circuit current.
   c) Current through the 2kΩ resistor.

6. A burglar alarm has 2 buzzers connected in parallel, which have a resistance of 200Ω and 300Ω and are connected to an 8v supply voltage. Calculate;
   a) The circuit resistance.
   b) The circuit current.
   c) The current through the 200Ω buzzer.

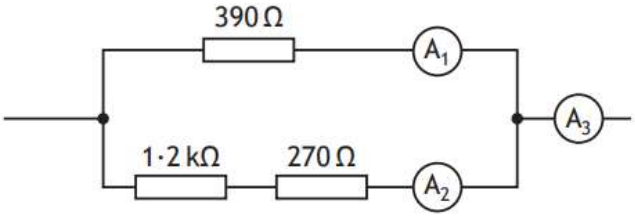
7. A student needs a 50Ω resistance to complete her electronics project. Unfortunately, she only has two 100Ω available. Draw a solution to her problem and explain your answer by providing proof (calculation) that it is correct.

8. A circuit with a voltage supply of 6v has a 30Ω and a 20Ω resistor connected in parallel and then they are connected in series to a 10Ω resistor. Draw the circuit and calculate;
   a) The total resistance of the circuit.
   b) The circuit current.
   c) The voltage dropped over the 10Ω resistor.
   d) The current through the 30Ω resistor.

9. **N5 past paper 2019 Q8.** An interactive speaker used in a 'smart home' is shown.



Part of the circuit used in the speaker is shown below.



a) Calculate the total resistance of this circuit — 3 marks

b) The reading on ammeter A_1 is 0.031 A.

(i) Calculate the voltage across the 390 Ω resistor. — 2 marks

(ii) Calculate the current A_2. — 3 marks

(iii) Calculate the current A_3. — 1 mark